

ENGR 1451/2451: Exploratory Data Science

Simple Linear Regression II
Residuals and Fit Diagnostics

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Today: What Does the Line Leave Unexplained?

Residuals

Which observations are above or below their fitted values?

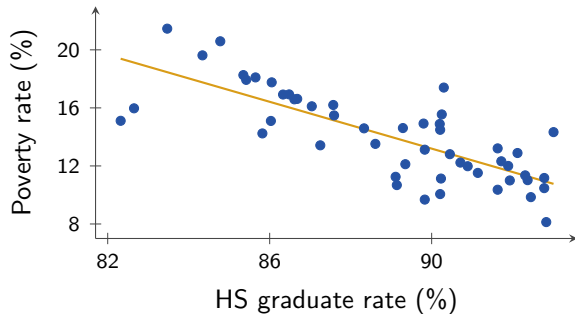
Size of the misses

How large are the residuals, and how much does the line improve on predicting the mean?

Diagnostics

Does the line miss a curve, a change in spread, a non-Gaussian shape, or an influential observation?

Continue with the previous Example



One observation

One of the 50 US states or DC.

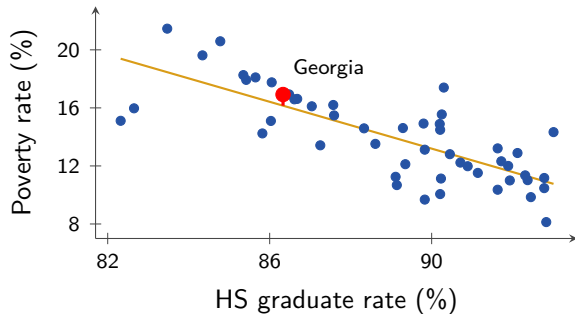
Same fitted line

$$\hat{y} \approx 85.789 - 0.806625x$$

Both variables are percentages.
These are **regional aggregates**, not individual residents.

2017 population-weighted county averages approximate statewide percentages.

Observed = Fitted + Residual



A signed vertical difference

$$e_i = y_i - \hat{y}_i$$

Georgia: rounded display values

Observed: 16.92%

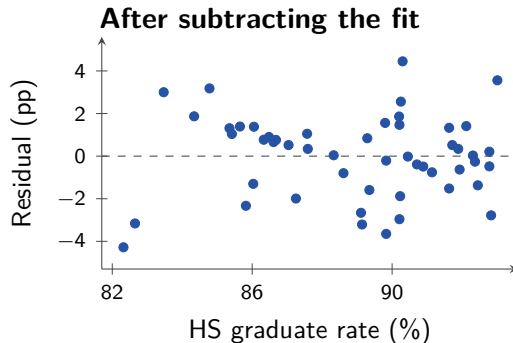
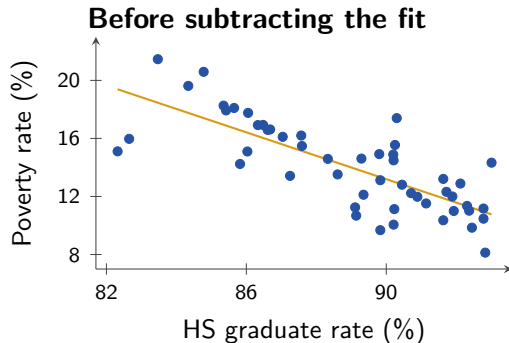
Fitted: 16.15%

Residual: +0.77 percentage points.

Positive: model underpredicts.

Negative: model overpredicts.

A Residual Plot Removes the Fitted Trend



Why?

Residuals may reveal a curved pattern, changing spread, or unusual points.

RMSE

Root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i e_i^2}$$

Square: positive and negative residuals cannot cancel.

Average: include all fitted observations.

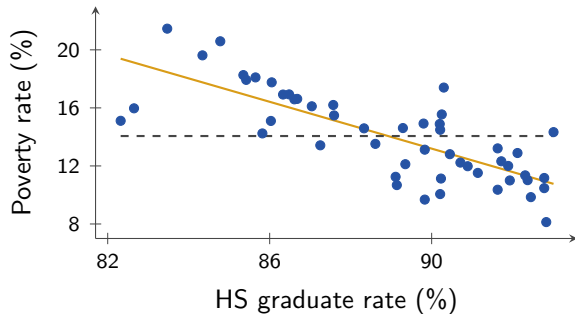
Square root: return to the response units.

In our example

$$\text{RMSE} \approx 1.90 \text{ pp}$$

A measure of typical residual size.

How Much Does the Line Improve on the Mean?



A simple baseline

The dashed line predicts $\bar{y}$ for every region, ignoring HS graduate rate.

Residual size in this example

Mean-only RMSE	3.00 pp
Line RMSE	1.90 pp

The fitted line reduces error on the **same data used to fit it**.

How much of the baseline *squared* error is removed?

R^2 Describes Improvement Relative to the Baseline

Compare squared error

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$$

$$\text{SSE} = \sum_i (y_i - \hat{y}_i)^2$$

$$\text{SST} = \sum_i (y_i - \bar{y})^2$$

Fraction of the mean-only squared error removed by the line, on these fitting data.

Here: $R^2 \approx 0.599$

About **60% of the variation in poverty rates across regions** is accounted for by the fitted line.

It is not percent correct and not a causal attribution.

For simple OLS with an intercept:
 $R^2 = r^2$. Here $r \approx -0.774$; only r retains the direction.

Check the Center, Spread, and Shape

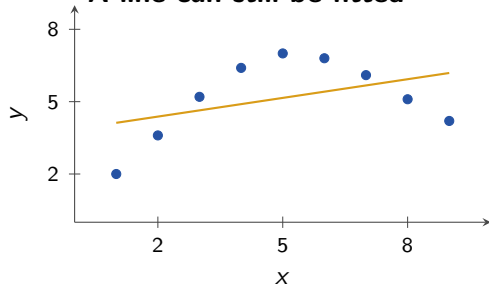
Question	What to inspect	Warning sign
Center	Scatterplot and residuals versus x	A curved residual pattern
Spread	Residuals versus x or fitted values	A fan or funnel
Shape	Residual histogram; normal probability plot	Skewness, heavy tails, isolated values

Different questions require different plots

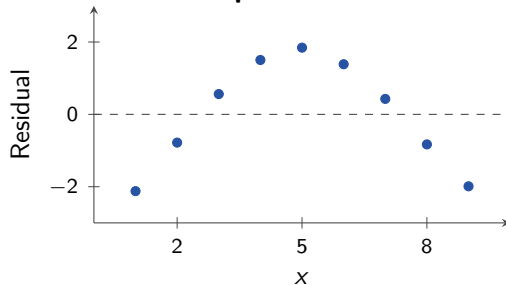
A bell-shaped histogram cannot show whether spread changes across the predictor range.
A large R^2 cannot substitute for any of these checks.

Check the Center: Residual Curvature

A line can still be fitted



The missed shape becomes visible

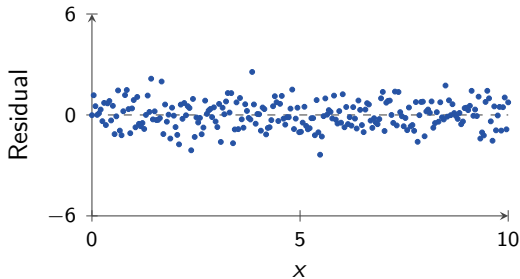


Interpret the pattern

Negative residuals at the ends and positive residuals in the middle indicate a curved average relationship that the line misses.

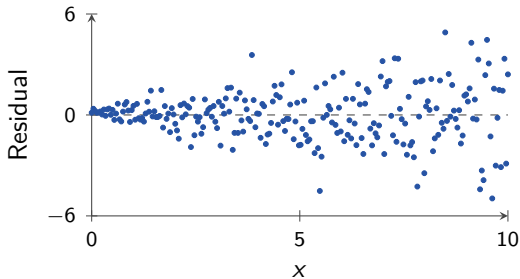
Homoscedastic or Heteroscedastic?

Roughly constant spread



Homoscedasticity

Changing spread: a fan



Heteroscedasticity

Same vertical scale. The key is the **vertical spread around zero**, not the slope of the original data.

Why Changing Spread Matters

One RMSE hides location-specific differences

Predictor range	n	RMSE
Low x	80	0.65
Middle x	80	1.39
High x	80	2.18

Overall simulated RMSE: 1.54 response units.

These equal-count groups summarize the fan-shaped example.

A line can describe the mean but not the spread

The average trend can be straight even when high- x observations have much larger residuals.

Investigate why variation changes. A more suitable model may describe spread as well as the mean.

Nearly Normal Means Gaussian-Like

An error model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \mid x_i \sim N(0, \sigma^2)$$

Equivalently,

$$Y_i \mid x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2).$$

At a fixed x , outcomes vary around the mean regression line. Under this model, the errors have a symmetric, bell-shaped distribution.

What we actually observe

After fitting the line,

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i,$$

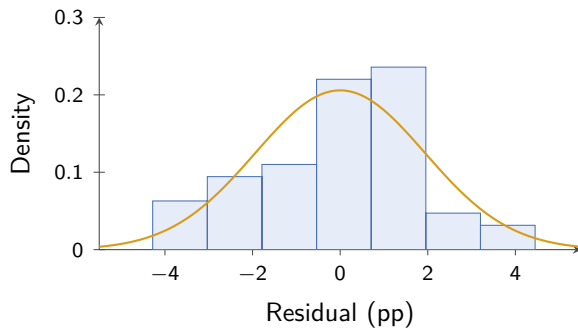
we calculate the **residual**

$$e_i = y_i - \hat{y}_i.$$

Residuals are observable estimates of the unobservable errors ε_i .

Gaussian errors are an additional modeling approximation.

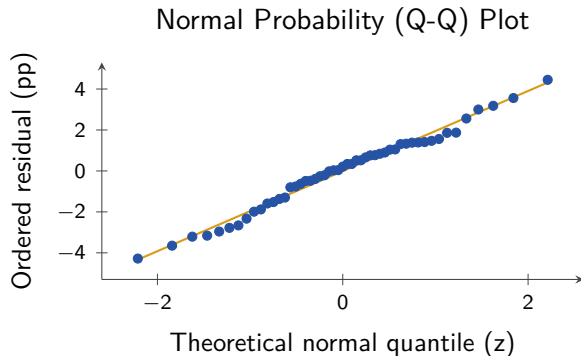
Check the Shape: A Residual Histogram



Look for approximate shape

A roughly symmetric, single-peaked distribution is compatible with a Gaussian-like error model.

Another Shape Check: A Normal Probability Plot



How to build the plot

Sort the residuals and pair each rank with the corresponding z-quantile from a standard normal distribution.

x-axis: Theoretical normal quantile (z)
y-axis: observed residual

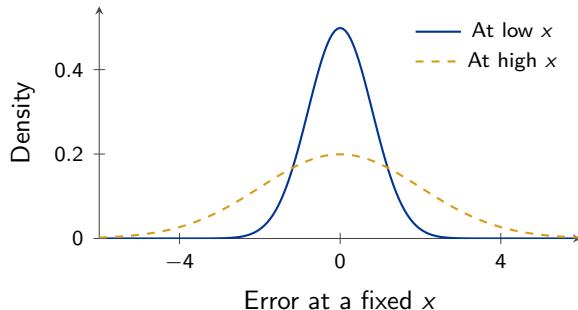
Why a straight line?

For approximately Gaussian residuals,

$$e \approx \mu_e + \sigma_e Z,$$

which is a linear relationship.

Gaussian Shape Is Not Constant Spread



Both curves are Gaussian

Both are centered at zero. Their standard deviations differ.

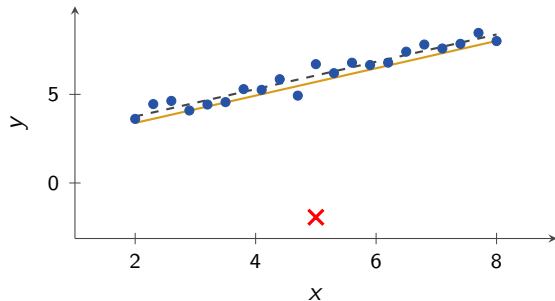
$$\varepsilon | x \sim N(0, \sigma(x)^2)$$

This is Gaussian-shaped noise **with heteroscedasticity**.

Three Ways an Observation Can Be Unusual

Feature	Question	Diagnostic
Large residual	Is y far from its fitted value?	Vertical distance from the line
High leverage	Is x far from the center of the predictor values?	Horizontal position
Influence on slope	Does omitting the point change the slope appreciably?	Refit with and without it

A Large Residual Need Not Change the Slope



Compare the fitted slopes

Without added point: 0.771

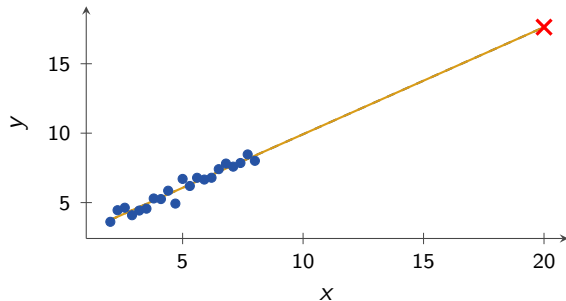
With added point: 0.771

The added point is near the center of the x values but far below the trend.

Here the slope does not change, but the **intercept and fitted values do**.

Dashed: fit without the added point. Solid: fit with it. Cross: added point.

High Leverage Need Not Mean Influence



Compare the fitted slopes

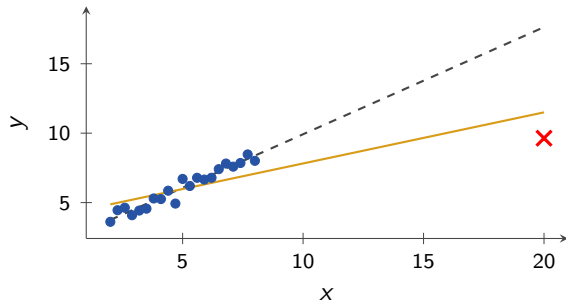
Without added point: 0.771

With added point: 0.771

The point is far away horizontally, but lies on the existing fitted trend.

It has high leverage yet changes neither fitted coefficient in this constructed example.

One Point Can Change the Slope



Compare the fitted slopes

Without added point: 0.771

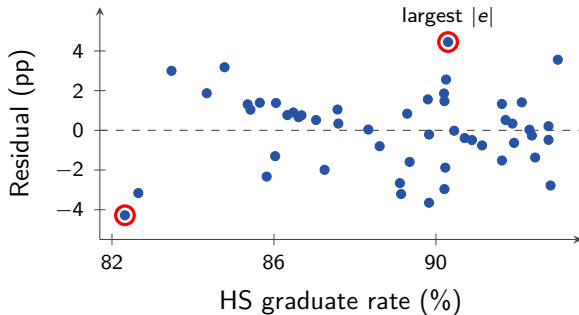
With added point: 0.368

The point is far away horizontally
and off the original trend.

It pulls the line toward itself. Its residual after fitting may understate how unusual it was relative to the original cloud.

Dashed: fit without the added point. Solid: fit with it. Cross: added point.

Back to the State Data: Investigate, Then Refit



Two different rankings

Which region has the largest $|e_i|$?

Which region changes the slope most when omitted?

They need not be the same region.

The notebook checks **every region in turn**, rather than assuming that the largest residual is the most influential point.

Sensitivity Is Not Permission to Delete

First: investigate

Check the source, units, data entry, population, and context. Is the observation a valid member of the study?

Then: compare

Display the original fit and the refit on the same predictor range. Examine changes in slope and fitted values.

Report, rather than conceal

Keep valid observations unless exclusion has a defensible reason. Explain how much conclusions depend on them.

Do not remove points just to increase R^2 , lower RMSE, or make a histogram look normal.

Which Problem Does Each Display Reveal?

A curved residual pattern

Does the line adequately represent the average relationship?

A fan-shaped residual plot

Is one common residual scale a good description across x ?

A bell-shaped histogram

Does this alone establish a good straight-line model?

Report the Model, Not Just a Number

Describe the fit

For these regional aggregates, a one-point higher HS graduate rate is associated with about a 0.81-point lower fitted poverty rate. RMSE is about 1.90 pp and R^2 about 0.60.

Describe the diagnostics

Comment separately on residual curvature, changing spread, Gaussian-like shape, and sensitivity to unusual observations.

State the limits

These are in-sample descriptions of regional averages. They do not establish an intervention effect, individual risk, or future prediction accuracy.

Takeaways: Read What the Line Leaves Behind

1. **Residuals:** signed misses in response units.
2. **RMSE and R^2 :** error size versus improvement on a mean-only baseline.
3. **Diagnostics:** check the center, spread, and shape separately.
4. **Unusual observations:** distinguish residual size, leverage, and influence.
5. **Context:** investigate observations and state the model's limits.